

Rasagyna Peddapalli, Rehan Ali, Utkarsh Choudhary, Yashwanth Nanjuti

Professor Syed Jawad Shah

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Customer Churn Prediction Using Machine Learning and Generative AI

1. Introduction

Customer churn, the voluntary termination of a service subscription, is among the most operationally significant challenges in telecommunications. Retaining a subscriber costs an estimated five to twenty-five times less than acquiring a replacement, making predictive churn modelling a commercially critical capability (Reichheld and Sasser 105). As global telecom markets approach saturation, operators must shift strategic emphasis from acquisition toward data-driven retention.

This project develops a four-phase machine learning and generative AI pipeline for churn prediction using the IBM Telco Customer Churn dataset, which contains 7,043 subscriber records across 21 attributes describing demographics, account tenure, subscribed services, and billing arrangements. The pipeline progresses through data ingestion and inspection, exploratory data analysis and preprocessing, supervised binary classification using a Random Forest compared against a Decision Tree baseline, and generative AI augmentation via BERT-based semantic embeddings, rule-based sentiment classification, an interactive Gradio interface, and a Streamlit dashboard with Plotly visualisations. The primary objective is a reliable classifier identifying at-risk customers prior to churn; the secondary objective is translating model outputs into interpretable natural-language summaries accessible to non-technical business stakeholders.

2. Related Work

Customer churn prediction has attracted sustained research interest since the early 2000s. Breiman introduced Random Forest in 2001, establishing bootstrap aggregation of decision trees as a dominant paradigm for tabular binary classification, valued for its robustness to overfitting and capacity to model non-linear feature interactions without explicit engineering (Breiman 5). Gradient boosting frameworks, including XGBoost (Chen and Guestrin 785) and LightGBM (Ke et al. 3146), subsequently demonstrated competitive accuracy through iterative residual correction.

A persistent challenge is class imbalance: telecom churn rates of 15 to 30 percent bias classifiers toward majority-class predictions. Chawla et al. introduced SMOTE to mitigate this through synthetic minority oversampling, substantially improving minority-class recall (Chawla et al. 321).

More recent work has extended churn analytics to natural language processing. Devlin et al. introduced BERT in 2019, a bidirectional transformer pre-trained through masked language modelling that achieves state-of-the-art results across language understanding benchmarks (Devlin et al. 4172). Applied to customer data, BERT enables contextual embeddings from narrative descriptions, capturing cross-feature semantics unavailable in flat tabular encodings. The integration of language models with structured churn classifiers has been identified as promising for improving minority-class recall and generating actionable retention signals (Abreu et al. 1). Deployment frameworks such as Streamlit and Gradio have further reduced the barrier between model development and business adoption.

3. Methodology

3.1 Phase 1: Data Ingestion and Inspection

The IBM Telco Customer Churn CSV file was loaded into a Pandas DataFrame. Schema inspection via `df.info()` recorded column names, data types, and non-null counts across all 21 fields. Descriptive statistics via `df.describe()` examined the continuous features: tenure, MonthlyCharges, and TotalCharges. The inspected dataset was persisted to a staging CSV file for reproducible downstream use.

3.2 Phase 2: Exploratory Data Analysis and Preprocessing

TotalCharges contained 11 whitespace entries for zero-tenure customers that caused object-type storage. These were coerced to numeric using `pd.to_numeric` with errors set to coerce, and the 11 resulting NaN rows were removed via `dropna`, leaving 7,032 records. The customerID column was excluded as a non-predictive identifier. Binary encoding mapped five Yes/No columns (Partner, Dependents, PhoneService, PaperlessBilling, Churn) and gender (Female = 1, Male = 0) to integers. One-hot encoding via `pd.get_dummies` with `drop_first` set to True expanded ten multi-category columns including MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies, Contract, and PaymentMethod.

Feature scaling was omitted, as Random Forest is invariant to input magnitude. The final encoded dataset comprised 7,032 records and 31 features.

3.3 Phase 3: Model Training and Evaluation

Data were partitioned using `train_test_split` at `test_size` equal to 0.20 with `random_state` fixed at 42, yielding 5,625 training and 1,407 test rows. A `RandomForestClassifier` with `n_estimators` equal to 100 and `random_state` equal to 42 was trained on the training partition. A `DecisionTreeClassifier` with `max_depth` equal to 10 was trained on identical partitions as a controlled baseline: since Random Forest constructs predictions by training 100 trees on different bootstrap samples and combining votes through majority voting, this pairing directly isolates the performance contribution of the ensemble mechanism. Both models were evaluated using accuracy score, a full per-class classification report, a confusion matrix via `ConfusionMatrixDisplay`, and ROC-AUC computed from `predict_proba` outputs. The trained Random Forest was serialised using `joblib` for Phase 4.

3.4 Phase 4: Generative AI and Semantic Analysis

The serialised model was reloaded in a separate notebook. A narrative function converted each record into a natural-language sentence encoding tenure (Long-term client if tenure exceeded 24 months, New client otherwise), charge tier (High-paying if `MonthlyCharges` exceeded 70), and contract type. The `bert-base-uncased` checkpoint was loaded from Hugging Face Transformers and 768-dimensional CLS token embeddings were extracted within `torch.no_grad` to suppress gradient computation during inference. A rule-based classifier assigned Neutral, Negative, or Frustrated labels by contract type and charge level. A Gradio interface enabled interactive risk assessment. The Streamlit dashboard provided Plotly-rendered overview charts, a confusion matrix, feature importances, and live single-customer scoring with `class_weight` set to `balanced` to address training imbalance.

4. Results and Discussion

4.1 Dataset Characteristics

After preprocessing, the dataset contained 7,032 records: 5,163 non-churners (73.46%) and 1,869 churners (26.54%), confirming moderate class imbalance. Table 1 summarises structural changes between the raw and processed datasets.

Table 1: Dataset Summary Before and After Preprocessing

Attribute	Raw Dataset	Processed Dataset
Records	7,043	7,032
Features	21	31 (after encoding)
Churn Rate	approx. 26.5%	26.54% (1,869 / 7,032)
Missing Values	11 rows in TotalCharges	Removed; 0 remaining
Train / Test Split	N/A	80/20: 5,625 / 1,407

EDA revealed pronounced churn variation across three dimensions. Contract type was most discriminative: month-to-month subscribers churned at 42.7% versus 11.3% for one-year and 2.8% for two-year contracts. Internet service type produced comparable divergence: fibre optic subscribers churned at 41.9% against 19.0% for DSL and 7.4% for no internet. Tenure exhibited a strong negative relationship: churn rates of 47.7% for customers under 12 months declined to 9.5% beyond 48 months. Mean tenure was 32.37 months, mean MonthlyCharges was 64.76, and mean TotalCharges was 2,283.30.

4.2 Model Performance

Table 2 compares both models. Decision Tree was selected as the baseline because it is the constituent base learner of Random Forest; comparing them on identical partitions directly quantifies the contribution of bootstrap aggregation while controlling for all other algorithmic variables.

Table 2: Model Comparison (Test Set, n = 1,407)

Model	Accuracy	ROC-AUC	Churn Recall	Churn F1
Decision Tree	74.27%	0.7363	0.5267	0.5212
Random Forest	78.75%	0.8184	0.4786	0.5449

Random Forest outperformed the Decision Tree across all four metrics. The ROC-AUC gap of 0.082 is the primary differentiating measure, evaluating discriminative ability across all thresholds rather than only the default 0.5 cutoff. Table 3 presents the full Random Forest classification report.

Table 3: Random Forest Classification Report

Class	Precision	Recall	F1-Score	Support
No Churn (0)	0.83	0.90	0.86	1,033
Churn (1)	0.63	0.48	0.54	374
Macro Avg	0.73	0.69	0.70	1,407
Weighted Avg	0.77	0.79	0.78	1,407

The classifier performed substantially better on the non-churn class ($F1 = 0.86$, $Recall = 0.90$) than the churn class ($F1 = 0.54$, $Recall = 0.48$). A churn recall of 0.48 means 195 of 374 actual churners were misclassified as non-churners, a 52.1% miss rate; each false negative represents a subscriber who departed without retention intervention. The confusion matrix confirmed 929 true negatives, 104 false positives, 195 false negatives, and 179 true positives. The root cause is class imbalance: 73.46% of training observations belong to the non-churn class, biasing predictions toward the majority. The Decision Tree achieved marginally higher churn recall (0.5267) but at the cost of 81 additional false positives and substantially lower ROC-AUC, confirming Random Forest as the superior model overall.

4.3 Generative AI and Semantic Analysis

BERT embeddings produced 768-dimensional contextual representations encoding cross-feature interactions as continuous geometric relationships unavailable in flat tabular encodings. The three sentiment tiers aligned directly with EDA findings: Neutral customers on long-term low-charge contracts correspond to the 2.8% churn rate for two-year subscribers; Negative month-to-month customers correspond to the 42.7% rate; and Frustrated customers combining month-to-month contracts with high charges correspond to the compounded risk reflected in the 41.9% fibre optic churn rate. The Streamlit dashboard operationalised outputs through four panels covering data overview, model evaluation, feature importances, and live risk scoring, bridging the gap between model development and non-technical business adoption.

4.4 Limitations

Class imbalance was not corrected in the notebook pipeline through resampling, directly suppressing churn recall. No hyperparameter optimisation was conducted, likely yielding a suboptimal minority-class boundary. BERT embeddings were derived from rule-constructed narratives rather than authentic customer feedback, limiting semantic richness. Evaluation relied

on a single train-test split without cross-validation, so reported metrics may vary across different random seeds.

5. Conclusion and Future Work

This project presented a complete four-phase pipeline for customer churn prediction. The Random Forest classifier achieved 78.75% accuracy and ROC-AUC of 0.8184 on the 1,407-record test set, outperforming the Decision Tree baseline by 4.48 percentage points in accuracy and 0.082 in ROC-AUC, empirically confirming the benefit of bootstrap aggregation. Churn recall of 0.48, attributable to the 73.46/26.54 class imbalance, was identified as the primary operational limitation, resulting in 195 actual churners receiving no retention intervention. BERT embeddings and rule-based sentiment classification produced three operationally differentiated customer profiles, Neutral, Negative, and Frustrated, mapped to differentiated retention strategies and aligned with the empirical churn rates observed in the exploratory analysis. The Gradio and Streamlit interfaces operationalised these outputs for non-technical users.

Future work should prioritise: applying SMOTE or `class_weight='balanced'` in the notebook pipeline to improve churn recall; systematic hyperparameter optimisation via GridSearchCV or Optuna targeting churn-class F1; benchmarking XGBoost and LightGBM against the Random Forest; replacing rule-constructed narratives with authentic customer support text to realise BERT's full representational capacity; fine-tuning BERT on a telecommunications corpus; and deploying the model as a FastAPI REST endpoint integrated with a live CRM system for real-time churn alerts at scale.

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